



SRI RAMAKRISHNA COLLEGE OF ARTS AND SCIENCE FOR WOMEN

CAREFLOW – SMART PATIENT APPOINTMENT & WAITING TIME MANAGEMENT SYSTEM

TEAM LEADER: GAYATHRISREE P

TEAM MEMBERS: MATHIMITHRA P R, INAS FATHIMA C, ANN JYOTSNA S,
JAYA PRIYA, KARTHINI G

FACULTY MENTOR: DR .D ANITHA

PROBLEM STATEMENT

Hospitals frequently experience prolonged and unpredictable patient waiting times as a result of inefficient appointment scheduling systems. This inefficiency leads to overcrowded waiting areas, causing discomfort and inconvenience for patients. Inadequate communication among doctors, patients, and hospital staff further aggravates the problem. Due to the absence of real-time schedule updates, appointments are often missed, delayed, or overlap. Consequently, patient dissatisfaction increases, and overall hospital efficiency is significantly reduced.

EXISTING SYSTEM

Existing appointment and queue management systems such as Practo and Qwaiting provide basic online appointment booking facilities. They rely on static or average-based waiting time calculations and mostly use manual methods to handle doctor delays. Patient communication is primarily through mobile applications or SMS notifications. These systems follow form-based interactions, offer limited multilingual support, and provide partial real-time updates regarding appointments and waiting status.

Drawbacks:

- Inaccurate and unpredictable waiting times due to static scheduling
- Late communication of doctor delays causing appointment issues
- Delayed and generic patient notifications
- Lack of two-way patient interaction
- Poor accessibility, especially for elderly patients
- Low transparency and weak hospital crowd control

PROPOSED SYSTEM

CareFlow is a smart, WhatsApp-integrated appointment and waiting time management system designed to digitize and optimize patient flow. The WhatsApp functionality is powered through the Gupshup API, and the app's backend and integrations were developed using Visual Studio Code. This setup allows real-time communication with patients, automated appointment reminders, and efficient management of waiting times, all hosted on a cloud-based infrastructure for scalability and reliability.

Working principle:

- Patient books an appointment through the CareFlow app
- System assigns a dynamic queue number instead of a fixed time slot
- Doctor availability and consultation start are tracked in real time
- Each patient's consultation duration is monitored continuously
- The system calculates live expected waiting time for all patients
- Patients receive alerts when their turn is approaching
- Delays or cancellations trigger automatic queue adjustment
- Emergency and priority patients are handled through override logic
- Patient visit status and history are updated after consultation

Features:

- Online appointment booking
- Real-time queue status
- Average waiting time prediction
- SMS notifications(Offline Booking)
- Doctor availability tracking
- Smart Prescriptions and reports(EMR integration)
- AI voice integration
- Hospital Navigation Assistant
- Smart rescheduling and priority handling
- Micro Consultant Service(Fast Lane)
- Medicine Delivery
- Multi-lingual capable

PROPOSED SYSTEM VS EXISTING SYSTEM

Aspect	Existing System	Proposed System
Appointment Booking	Available	Available
Waiting Time Logic	Static / average-based	Real-time, dynamic
Delay Handling	Manual / delayed	Automatic detection
Patient Notifications	Delayed	Immediate
Communication Mode	App / SMS	WhatsApp
Patient Interaction	One-way	Two-way
Accessibility	Limited, not elderly-friendly	High, voice & user-friendly
Transparency	Low	High
Hospital Crowd Control	Weak	Strong
Overall Experience	Uncertain	Predictable & stress-free

FEASIBILITY STUDY

Technical Feasibility

- Uses standard mobile and web technologies
- Cloud database and real-time APIs
- Easy integration with existing hospital workflows

Economic Feasibility

- Reduces operational inefficiencies
- Saves staff time and resources

Operational Feasibility

- User-friendly interface for patients and staff
- Suitable for both small clinics and large hospitals

Social Feasibility

- Reduces patient stress and frustration
- Improves doctor–patient interaction

Financial Feasibility

- App-based solution with minimal hardware dependency
- Low development and maintenance cost
- Scalable across clinics, hospitals, and diagnostic centers
- Cloud-based backend for real-time updates
- Estimated cost: ₹ 70,000-80,000

IMPACT

- Reduced patient waiting time
- Improved doctor utilization
- Decongested waiting areas
- Higher patient satisfaction

IDEAS REVISED BASED ON JURY FEEDBACK

- QR based appointment
- Voice-based booking
- Average waiting time prediction

CONCLUSION AND FUTURE ENHANCEMENTS

CareFlow is not merely an appointment booking application but a comprehensive smart patient flow management solution. By directly addressing challenges in appointment scheduling and waiting time management, it enhances hospital efficiency, minimizes patient stress, and delivers a more transparent and predictable healthcare experience. Through real-time updates, intelligent delay handling, and patient-friendly communication, CareFlow bridges the gap between hospitals and patients, leading to improved satisfaction and smoother operations.

In the future,

- Virtual Consultations
- Internet failure: Offline token sync
- Voice output
- Lightweight app version ,will be added

WORKFLOW:

https://drive.google.com/file/d/1egbsJcQ5i_mgwaTEQjF65bEbRKA2rtfE/view?usp=sharing